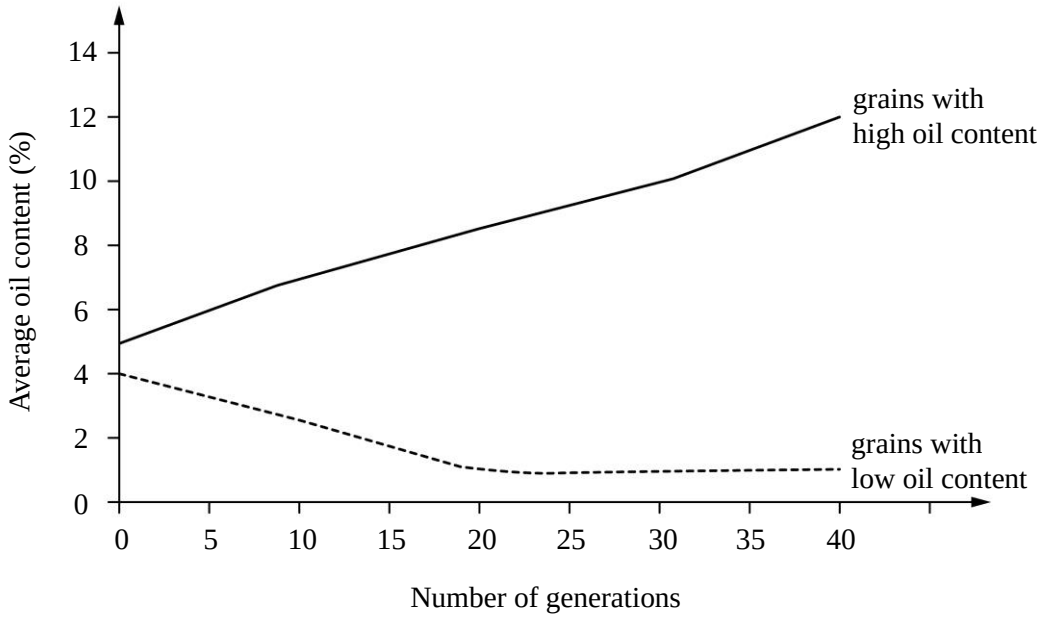
Short questions

|!|BCESB0310000001|!| ★

Two groups of researchers tried to alter the oil content of maize grains by selective breeding. Group A selected only grains with high oil content to cultivate, while group B selected and grew only grains with low oil content. This selection process was repeated for 40 generations. The oil content of the grains in each generation was measured and the results are shown in the graph below.



(a) (i) For group A, in which generation had the oil content of the maize grains been doubled? (1 mark)

_____

(ii) For group B, in which generation had the oil content of the maize grains been halved? (1 mark)

_____

(b) The difference in the average oil contents between the two groups becomes larger as the number of generations increases. Considering the 40th generation, find the difference in the average oil content between the two groups. (1 mark)

_____

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(c) Which group of maize grains would have a greater population if they grew in the same natural environment? Why? (2 marks)

##

(a) (i) 30th generation (1)

(ii) 13th generation (1)

(b) 11% (1)

(c) Grains with high oil content (1)

These grains have more energy reserve for germination and the growth of the young seedlings. (1)

(5 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)	Analysing data	1
(c)	Applying concepts	2

|!|BCESC0310000002|!|

The photograph below shows a black peppered moth on a tree trunk.



(a) Describe ***one*** favourable characteristic of the black peppered moth that enables it to adapt to its environment. (1 mark)

(b) Explain why the black peppered moth has a selective advantage over the white peppered moth on the tree trunk as shown in the photograph. (2 marks)

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##

(a) Its dark-coloured body is well-camouflaged on blackened tree trunks. (1)

(b) On blackened tree trunks, the black peppered moth is hardly spotted and is less likely to be eaten by birds. (1)

The black peppered moth has a greater chance to survive and reproduce. (1)

The white peppered moth is not camouflaged on blackened tree trunks and is more likely to be eaten by birds. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3

|!|BCESC0310000003|!|

Mutations in genes can produce new traits in individuals and may lead to the formation of new species.

(a) Give an example of mutagens that leads to mutation. (1 mark)

(b) In terms of natural selection, explain how mutation contributes to the evolution of different finches on the Galapagos Islands. (5 marks)

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##

(a) Ionizing radiation / ultraviolet light / other reasonable answer (1)

(b) There was a lack of seeds for the ancestral finches which first arrived on the Galapagos Islands. (1)

Genetic variations exist among the finches due to random mutations. (1)

Some finches had slightly different beaks which enabled them to feed on foods (e.g. insects) other than seeds. (1)

These finches had a greater chance to survive, reproduce and pass the favourable characteristics to offspring. (1)

This happened over many generations and the favourable variations accumulated in the population. (1)

The population of finches has evolved and become different from its ancestor.

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	5

|!|BCESC0310000004|!|

Complete the following paragraph with suitable words. (5 marks)

Organisms tend to produce more offspring than the environment can support, hence individuals compete for (a)_____ (e.g. food, water and shelter), i.e. there is a (b)_____ . Individuals with (c)_____ favourable to their environment have a greater chance to survive and reproduce. Other less well adapted individuals fail to survive or reproduce. This is known as (d)_____. Through many generations, the variations accumulate and individuals with the favourable characters become more (e)_____ in the population.

##

(a) limited resources (1)

(b) struggle for existence (1)

(c) characteristics / variation (1)

(d) survival of the fittest (1)

(e) frequent (1)

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_____
(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESC0310000005|!|

Complete the following table which lists two examples of natural selection. (7 marks)

	A population of mosquitoes in an environment sprayed with pesticides	**A population of giraffes**
Variation within the species	(a)_____ mosquitoes and (b)_____ mosquitoes	Short-necked giraffes and long-necked giraffes
Selection pressure	Application of pesticides to eliminate mosquitoes	Shortage of food at the ground level but surplus of food high up in trees
Form selected for	(c)_____ mosquitoes	(e)_____ giraffes
Form selected against	(d)_____ mosquitoes	(f)_____ giraffes
Effect of natural selection over many generations	The proportion of the form selected for in the population (g) increases / decreases , while the proportion of the form selected against in the population (h) increases / decreases .	

##

(a) pesticide-resistant / non-resistant (1)
(b) non-resistant / pesticide-resistant (1)
(c) pesticide-resistant (1)
(d) non-resistant (1)
(e) long-necked (1)

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(f) short-necked (1)
(g) increases ($^1/_2$)
(h) decreases ($^1/_2$)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1
(g)	Understanding basic concepts	$^1/_2$
(h)	Understanding basic concepts	$^1/_2$

|!|BCESC0310000006|!|

Complete the following paragraph with suitable words. (5 marks)

A population of the same species was divided into two populations by a (a)_____, such as desert, river or mountain range. This is called (b)_____. Each separated group evolved independently by (c)_____. Over time, the two populations become so different that they could no longer interbreed. Each population becomes a new (d)_____. This process is called (e)_____.

##

(a) physical barrier (1)
(b) geographic isolation (1)
(c) natural selection (1)
(d) species (1)
(e) speciation (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESA0310000007|!| ★★

The common chimpanzee (*Pan troglodytes*) which lives north of the Congo River, and the bonobo (*Pan paniscus*) which lives south of it, were once considered to be the same species, but since 1928 they have been recognized as different species.

(a) Suggest how scientists could determine whether or not two organisms belong to the same species. (2 marks)

(b) Both the common chimpanzee and the bonobo are poor swimmers. The formation of the Congo River about 1.5 million years ago possibly led to the speciation of the common chimpanzee and the bonobo. Suggest how these two species might have evolved from their common ancestor. (5 marks)

##

(a) By conducting a cross between the two organisms (1)
Individuals of different species cannot interbreed and produce fertile offspring. (1)

(b) A population of the ancestors of the common chimpanzee and the bonobo were separated into isolated groups after the formation of the Congo River. (1)
Individuals of the isolated groups could not meet or breed with each other. There was no gene flow between the isolated groups. (1)
The environmental conditions on the north and south of the river were different. By natural selection, the favourable variations selected for in each isolated group were different. (1)

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Over many generations, variations accumulated in each isolated group. (1)
The isolated groups have become very different from each other and could no longer interbreed to produce fertile offspring. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	5

|!|BCESC0310000008|!|

Lamarck and Darwin proposed different explanations for the evolution of long necks in giraffes. Distinguish whether each of the ideas about evolution listed in ***Column 2*** was proposed by Lamarck or Darwin. Put the appropriate letters in the space provided.

Column 1

Darwin (a) _____

Lamarck (b) _____

Column 2

A. Giraffes stretched their necks during lifetime and hence their necks became longer and longer.
B. Characteristics acquired during lifetime can be passed on to offspring.
C. Variation exists among the giraffes, i.e. some giraffes were born with short necks and some with long necks.
D. Organisms change their characteristics in order to adapt to the environment.
E. The environment selects individuals with favourable characteristics to survive and reproduce.

##

(a) C, E (2)
(b) A, B, D (3)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3

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